Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Chuanqi Tan, and Chang Zhou. Scaling relationship on learning mathematical reasoning with large language models. *arXiv preprint arXiv:2308.01825*, 2023a.

Zheng Yuan, Hongyi Yuan, Chuanqi Tan, Wei Wang, Songfang Huang, and Fei Huang. Rrhf: Rank responses to align language models with human feedback without tears. *arXiv preprint arXiv:2304.05302*, 2023b.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023.

I want you to act as a grade school math teacher, and evaluate the quality of the answer provided by an AI assistant to the math Question displayed below.
You will be given a reference answer and the assistant's answer, and Your evaluation should consider the correctness of the assistant's answer.
Begin your evaluation by comparing the assistant's answer with the reference answer step-by-step. Identify and correct any mistakes.
The answer is scored out of 10 points, with one point deducted for each wrong step. Be as objective as possible.
Your need first provide your Evaluation Evidence and then rate the response on a scale of 1 to 10.
[Question]:
{question}
[The Start of Reference Answer]
{reference}
[The End of Reference Answer]
[The Start of Assistant's Answer]
{answer}
[The End of Assistant's Answer]
You MUST output with two lines:
Evaluation Evidence: <Explanation>
Rating: <ONLY a single digit>

Table 6: The evaluation template that prompts ChatGPT to score each candidate COT.

## A Datasets

We conduct our experiments on three widely used reasoning datasets with human-annotated chain-of-thoughts, including math reasoning tasks GSM8K (Cobbe et al., 2021), AQUA-RAT (Ling et al., 2017), commonsense reasoning task ECQA (Aggarwal et al., 2021):

**GSM8K** GSM8K is a widely used mathematical reasoning dataset, which comprises 8.5K varied math word problems for grade school, developed by human authors. It is partitioned into 7.5K training problems and 1K testing problems. We sample 400 problems from the testing set to form the validation set, and thus we have $7,473$, $400$, and $919$ examples in training, validation, and testing sets, respectively.

**AQUA-RAT** AQUA-RAT comprises approximately $100,000$ algebra-based word problems, each accompanied by a natural language rationale. Each example in the dataset consists of four components: 1) question, which statement is written in natural language, 2) options, a set of five potential answers with one being correct, 3) rationale, a natural language explanation of the problem's solution, and 4) correct, the right answer choice. For efficiency, we randomly sample $5,000$, $400$, and $1,254$ examples as the training, validation, and test set, respectively.

**ECQA** ECQA is derived from CommonsenseQA (CQA) (Saha et al., 2018) by generating a free-flow explanation for each QA pair in CQA. CQA is a comprehensive dataset for commonsense reasoning, containing QA pairs with five choices and a single correct answer. ECQA comprises 11K QA pairs in total and has $7,598$, $1,090$, and $2,194$ examples in the training, validation, and test sets, respectively.

**GSM8K-RANK** To evaluate the effectiveness of our AFT in the ranking situation, we randomly select 1,000 examples from GSM8K's training set and generate 8 candidate COTs for each question. We then prompt ChatGPT to rate these candidates by providing the question, reference answer, and the COT to be assessed and thus we can achieve a quality ranking sequence for different generated COTs. We randomly sampled 20 examples and found that ChatGPT's scoring results align well with human assessment. ChatGPT is instructed to assign a score between 1 and 10, indicating the quality of each COT. To ensure the reliability of the ratings, following (Wang et al., 2023b), we require ChatGPT to present evaluation evidence before assigning a score, and simple 3 scores for each example. We take the average score as the final score for each COT.

| Models | GSM8K | AQUA | ECQA | GSM8K-RANK |
|---|---|---|---|---|
| LLama-7B | 0.15 | 0.15 | 0.15 | 0.05 |
| LLama2-7B | 0.15 | 0.40 | 0.35 | 0.15 |
| LLama-13B | 0.15 | 0.15 | 0.15 | 0.15 |
| LLama2-13B | 0.15 | 0.15 | 0.20 | 0.15 |

Table 7: The value of hyper-parameter $\beta$ for boundary constraint alignment.

| Methods | $\mathcal{L}_{VFT}$ | +$L_A^{RBC}$ | +$\mathcal{L}_A^{RDC1}$ | +$\mathcal{L}_A^{RDC2}$ | +$\mathcal{L}_A^{R}$ |
|---|---|---|---|---|---|
| Accuracy | 20.82±0.71 | 26.08±1.05 | 25.68±0.49 | 12.57±1.34 | 7.03±0.98 |

Table 8: Results of LLama-7B on GSM8K fin-tuned by different methods.

## B PARAMETER SETTING

We conduct experiments on four large language models, LLama-7B, LLama-13B, LLama2-7B, and LLama2-13B. We do not conduct experiments on larger models due to resource limitations. We sample $k = 6$ COTs from VFT-LLMs with a sampling temperature of 1. Our detached constraint alignment loss does not introduce any hyper-parameters, and we search the hyper-parameter of boundary constraint loss within the range $(0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5)$ on the validation set. The value of $\beta$ of different models and datasets is provided in Table 7. On GSM8K, AQUA, and ECQA, the models are trained for 3, 3, and 1 epochs, respectively. The learning rate is set to 2e-5, featuring linear decay and a linear warmup for 3% of the total training steps. 7B and 13B models are trained on 8 and 32 V100 GPUs with 32GB memory, respectively. We employ a maximum sequence length of 512 and utilize the DeepSpeed library and ZeRO optimizer during training.

## C DETACHED CONSTRAINT RANKING LOSS

Given a ranking sequence $c_1 \succeq c_2 \succeq \cdots \succeq c_k$, besides extending $\mathcal{L}_A^{BC}$ (Equation 8) to the ranking loss $L_A^{RBC}$ (Equation 14), we also try to extend $\mathcal{R}_A^{DC}$ to two types of detached constraint ranking loss as follows:

$$L_A^{RDC1} = \log\left[1 + \sum_{c_i \succ c_j} \exp(\mathbf{D}(s_\theta^{c_j}) - s_\theta^{c_i})\right] \tag{12}$$

$$L_A^{RDC2} = \log\left[1 + \sum_{c_i \succ c_j, c_j \notin c_{min}} \exp(s_\theta^{c_j} - s_\theta^{c_i}) + \sum_{c_i \succ c_j, c_j \in c_{min}} \exp(\mathbf{D}(s_\theta^{c_j}) - s_\theta^{c_i})\right] \tag{13}$$

where $c_{min}$ is the set of all lowest-quality examples. Specifically, $L_A^{RDC1}$ detachs the score of $c$ when it serves as a negative example, while $L_A^{RDC2}$ only detach the score of lowest-quality examples. We design $L_A^{RDC2}$ as we consider that in a ranking scenario, higher-quality examples are inherently constrained by lower-quality ones. Consequently, we hypothesize that constraining only the lowest examples could potentially prevent model degradation.

We also consider a ranking baseline without any constraint:

$$\mathcal{L}_A^{R} = \log\left[1 + \sum_{c_i \succ c_j} \exp(s_\theta^{c_j} - s_\theta^{c_i})\right] \tag{14}$$

Table 8 illustrates the results of LLama7B fine-tuned by different methods on GSM8K-RANK. As is shown: **1)**: The method without setting any constraint $\mathcal{L}_A$ only achieves 7.03 accuracy, showing